TASK 5 • FINAL TASK

AI & ML in the Real World

Five real-world problems across agriculture, healthcare, education, transportation and the environment — and how Artificial Intelligence & Machine Learning can solve them.



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Introduction & Objective

Artificial Intelligence and Machine Learning are no longer confined to research labs — they are actively reshaping how we grow food, deliver healthcare, teach students, move through cities, and protect our environment. This report identifies five real-world problems spanning agriculture, healthcare, education, transportation, and environment, and proposes practical AI/ML-based solutions for each, along with how they would work and the benefits they would bring.

OBJECTIVE

To develop practical thinking about the use of Artificial Intelligence and Machine Learning in solving genuine, everyday real-world problems — moving from theory to applied impact.

1. **Crop Disease Detection — Agriculture**
2. **Early Disease Diagnosis — Healthcare**
3. **Personalized Learning — Education**
4. **Traffic Congestion Management — Transportation**
5. **Disaster Prediction — Environment**

AGRICULTURE

01

Crop Disease Detection

THE PROBLEM

Farmers, especially in rural areas, often cannot identify crop diseases early. By the time symptoms are visible to the naked eye, significant yield loss has already occurred — and misdiagnosis frequently leads to the wrong pesticide being used, wasting money and harming soil health.

AI/ML SOLUTION

A Convolutional Neural Network (CNN)-based image classification model that identifies plant diseases directly from a leaf photo taken on an ordinary smartphone.

HOW IT WORKS

- Farmer photographs an affected leaf using a mobile app
- Image is fed into a CNN trained on labelled datasets (e.g. PlantVillage)
- Model classifies the disease — blight, rust, mildew, etc.
- App instantly suggests the correct treatment or pesticide dosage

EXPECTED BENEFIT

Early detection reduces crop loss, cuts pesticide misuse, increases yield, and is accessible even to farmers with only a basic smartphone and no expert nearby.

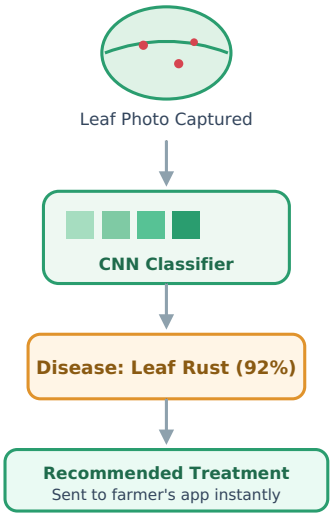


FIG 1.1 — CROP DISEASE DETECTION PIPELINE

AGRICULTURE · DEEPER DIVE

Crop Disease Detection

01

REAL-WORLD CASE STUDY

Plantix (PEAT GmbH)

Plantix is a free crop-advisory app that lets farmers photograph a leaf and instantly receive an AI-generated diagnosis, along with treatment suggestions and connections to local input sellers. It uses deep learning-based image recognition trained on a crowdsourced library of real farmer photos, which keeps improving with every new image submitted.

400+	10M+	100M+
PESTS & DISEASES DETECTED	APP DOWNLOADS	FARMER QUESTIONS ANSWERED

KEY TECHNOLOGIES USED

- Convolutional Neural Networks
- TensorFlow / PyTorch
- Mobile Image Capture
- Cloud Inference API
- Crowdsourced Datasets
- WhatsApp / Chatbot Integration

CHALLENGES TO CONSIDER

- ! Image quality varies with lighting, angle and camera — models must be trained to handle noisy, real-world photos.
- ! Rural areas often have weak internet connectivity, so offline or low-bandwidth modes are essential.
- ! Model accuracy must be continuously validated across different crops, regions and disease strains.

✓ **Why it matters:** Up to 30% of the world's annual harvest is lost to pests and disease. Putting an expert-level diagnostic tool in every farmer's pocket — even without an agronomist nearby — directly protects food security and rural livelihoods.

HEALTHCARE

02

Early Disease Diagnosis

THE PROBLEM

Diseases such as diabetic retinopathy, cancer, and heart disease often go undiagnosed until late stages because of limited access to specialists, particularly in rural or underserved regions.

AI/ML SOLUTION

Machine learning models — deep learning on medical imaging, or classification models on patient records — that flag high-risk cases for early, priority screening.

HOW IT WORKS

- Patient data (X-rays, retinal scans, blood reports) is collected
- Data is fed into a trained diagnostic model
- Model outputs a probability score for disease risk
- High-risk cases are flagged and routed to a doctor for review

EXPECTED BENEFIT

Faster diagnosis, reduced burden on specialists, earlier treatment, and better patient outcomes — especially valuable where doctors are scarce.

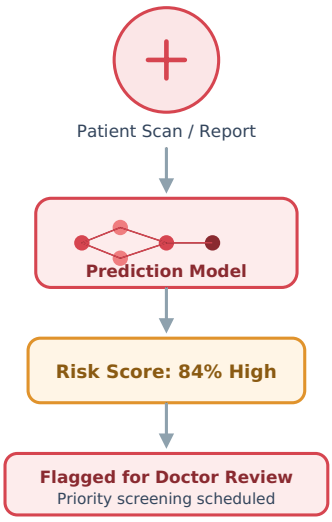


FIG 2.1 — EARLY-WARNING DIAGNOSTIC PIPELINE

Early Disease Diagnosis

REAL-WORLD CASE STUDY

Google's Diabetic Retinopathy Detection Model

Google Research trained a deep neural network on over 128,000 retinal fundus images, each graded multiple times by licensed ophthalmologists, to detect diabetic retinopathy — the leading cause of vision loss in diabetics. A later study found that ophthalmologists assisted by the AI's grading and heatmap explanations diagnosed more accurately than either the doctors or the AI working alone.

97.5%	93.4%	128K+
SENSITIVITY ACHIEVED	SPECIFICITY ACHIEVED	TRAINING IMAGES USED

KEY TECHNOLOGIES USED

- Deep Neural Networks
- Medical Image Processing
- Explainable AI (Heatmaps)
- Cloud-based Inference
- Electronic Health Records
- Clinical Validation Pipelines

CHALLENGES TO CONSIDER

- ! Patient data privacy and regulatory compliance (e.g. HIPAA-equivalent laws) are critical and non-negotiable.
- ! Models must be validated across diverse populations to avoid bias from unrepresentative training data.
- ! AI should assist, not replace, clinical judgment — final decisions still rest with qualified doctors.

✓ **Why it matters:** Screening at this level of accuracy, delivered near-instantly and at low cost, means high-risk patients in areas with few specialists can still be flagged early — when treatment is most effective.

EDUCATION

03

Personalized Learning

THE PROBLEM

Traditional classrooms follow a one-size-fits-all approach, but students learn at different paces and have different strengths and weaknesses — leading to disengagement or students falling behind.

AI/ML SOLUTION

An adaptive learning system that uses machine learning to personalize content difficulty and pacing for every individual student.

HOW IT WORKS

- System tracks quiz scores, time taken, and error patterns
- ML model identifies each student's weak topics
- Content difficulty and pacing adjust dynamically
- Targeted practice material is recommended automatically

EXPECTED BENEFIT

Improved learning outcomes, reduced dropout rates, and more efficient use of teacher time by automating basic assessment and remediation.

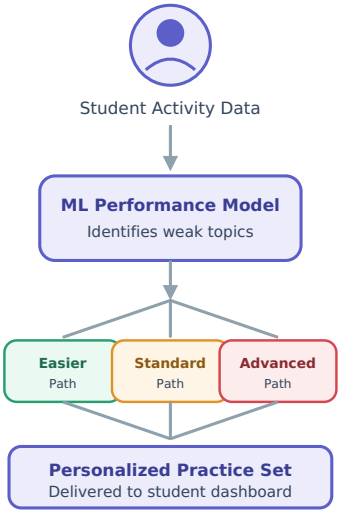


FIG 3.1 — ADAPTIVE LEARNING LOOP

EDUCATION · DEEPER DIVE

Personalized Learning

REAL-WORLD CASE STUDY

Squirrel AI Learning (China)

Squirrel AI breaks subjects like middle-school mathematics into over 10,000 fine-grained "knowledge points" linked in a knowledge graph, then uses AI to diagnose each student's gaps and decide what to teach next. In a widely reported "AI teacher vs human teacher" trial, students tutored by the adaptive system mastered more knowledge points in the same eight-hour window than those tutored by experienced human teachers.

24M+	10,000+	42 vs 28
REGISTERED STUDENTS	KNOWLEDGE POINTS MAPPED	POINTS LEARNED / 8 HRS (AI VS HUMAN)

KEY TECHNOLOGIES USED

- **Knowledge Graphs**
- **Bayesian Knowledge Tracing**
- **Adaptive Recommendation Engines**
- **Learning Analytics Dashboards**
- **Diagnostic Testing Models**
- **Large Adaptive Models (LAMs)**

CHALLENGES TO CONSIDER

- ! Needs a very large, well-labelled bank of practice questions mapped to fine-grained topics — expensive to build.
- ! Over-personalization can reduce collaborative learning and peer interaction if not balanced with classroom time.
- ! Requires devices and stable connectivity, which can widen gaps for students without reliable access.

✓ **Why it matters:** When every student gets a learning path matched to their actual gaps instead of the class average, weaker students catch up faster and stronger students aren't held back — at a scale no single teacher could manage alone.

TRANSPORTATION

04

Traffic Congestion Management

THE PROBLEM

Urban traffic congestion wastes time, fuel, and increases pollution, especially in cities where signals run on fixed timers regardless of actual traffic volume.

AI/ML SOLUTION

A reinforcement learning-based adaptive traffic signal control system that adjusts signal timing in real time based on live traffic density.

HOW IT WORKS

- Cameras and sensors feed live traffic data at each junction
- An RL agent learns optimal signal-timing policies
- Signals dynamically adjust green/red durations
- System continuously improves from ongoing traffic patterns

EXPECTED BENEFIT

Reduced travel time, lower fuel consumption, decreased emissions, and smoother traffic flow — especially during peak hours.

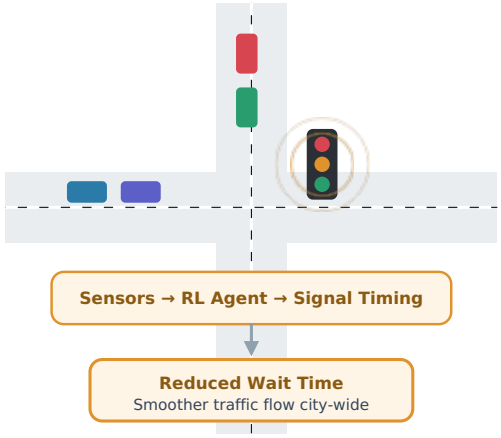


FIG 4.1 — ADAPTIVE TRAFFIC SIGNAL CONTROL

TRANSPORTATION · DEEPER DIVE

Traffic Congestion Management

REAL-WORLD CASE STUDY

Surtrac — Carnegie Mellon University, Pittsburgh

Developed by CMU's Robotics Institute and commercialized as Rapid Flow Technologies, Surtrac uses radar sensors and cameras at each intersection to build a real-time signal-timing plan, then shares projected outflows with neighboring lights so green corridors form automatically — with no central controller. First piloted on 9 intersections in 2012, it has since expanded to 50+ across the city.

25%	40%	21%
AVG. TRAVEL TIME CUT	AVG. WAIT TIME CUT	EMISSIONS REDUCED

KEY TECHNOLOGIES USED

- Reinforcement Learning
- Radar & Camera Sensors
- Decentralized Multi-Agent AI
- Real-Time Scheduling Algorithms
- Edge Computing at Intersections
- V2I Communication

CHALLENGES TO CONSIDER

- Hardware (sensors, edge computers) at every junction is a significant upfront infrastructure cost.
- Must safely handle pedestrians, cyclists and emergency vehicles, not just car throughput.
- City-wide coordination needs robust communication networks between neighboring intersections.

✓ **Why it matters:** Traffic congestion alone costs the U.S. economy an estimated $121 billion a year in lost productivity. Signals that think in real time turn static infrastructure into an adaptive system — without laying a single new road.

ENVIRONMENT

05

Disaster Prediction

THE PROBLEM

Natural disasters like floods and cyclones cause massive damage, and delayed or inaccurate warnings result in avoidable loss of life and property.

AI/ML SOLUTION

Machine learning models trained on historical weather and satellite data to predict floods, cyclones, or heatwaves with higher accuracy and greater lead time.

HOW IT WORKS

- Historical weather, satellite imagery, and live sensor data are gathered
- Time-series models (e.g. LSTM networks) analyze the patterns
- Model predicts likelihood and severity of an event
- Early alerts are issued to authorities and residents

EXPECTED BENEFIT

Earlier evacuation and preparation time, reduced casualties, and better resource allocation for disaster response teams.

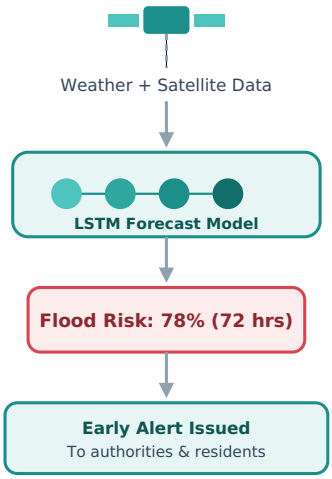


FIG 5.1 — DISASTER EARLY-WARNING PIPELINE

Disaster Prediction

REAL-WORLD CASE STUDY

Google Flood Hub

Flood Hub combines two AI models — a Hydrologic Model that forecasts how much water is flowing through a river, and an Inundation Model that predicts which areas will flood and how deep the water will get — using publicly available weather and satellite data. First piloted in India's Patna region in 2018, it now issues forecasts up to seven days in advance and sends free alerts directly to residents' phones.

150+	7 Days	460M+
COUNTRIES COVERED	ADVANCE WARNING	PEOPLE PROTECTED

KEY TECHNOLOGIES USED

- LSTM / Time-Series Models
- Satellite Imagery Analysis
- Hydrological Modeling
- Geospatial Mapping
- Push Notification Systems
- Public Weather Data APIs

CHALLENGES TO CONSIDER

- ! Alerts require internet-connected smartphones, which excludes some of the most at-risk, low-connectivity communities.
- ! Currently strongest for riverine floods — flash floods and other fast-onset disasters are harder to model.
- ! Needs partnerships with local governments and aid groups to turn a forecast into an actual evacuation.

✓ **Why it matters:** Floods affect over 250 million people and cause roughly $10 billion in damage every year worldwide. A week of advance notice is often the difference between an orderly evacuation and a disaster.

Conclusion

These five examples show a consistent pattern: AI/ML systems take in real-world data — images, sensor readings, patient records, or historical patterns — learn from it, and produce predictions or decisions that would be too slow or too complex for manual processes alone. Applied thoughtfully, this creates measurable benefits: less crop loss, faster diagnoses, better-personalized education, smoother cities, and fewer disaster casualties.

Of these, **Crop Disease Detection using CNNs** stands out as the idea with the most immediate, grassroots impact — it needs only a smartphone, works offline-friendly, and directly supports farmers who often have the least access to expert help.

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